

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY COURSE
CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS:
Annexure A

Total Marks: 50

Code of Conduct:

I Kartheeka Repalle (192372289) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Kartheeka Repalle

Annexure A

Scenario-based

1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

- A. Explain why this issue occurs.
- B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

- A. Identify the most suitable algorithm for this scenario.
- B. Justify your choice based on security strength and efficiency.

3. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

- A. Explain how the key exchange works in this scenario.
- B. Identify a possible security threat and suggest a solution to mitigate it.

4. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

- A. Describe how RSA can be used for encryption and decryption in this scenario.
- B. Explain how key distribution and management play a role in maintaining security.

5. Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

- A. Analyze the constraints of IoT devices.
- B. Recommend a suitable cryptographic approach and justify your answer.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Secure Data Transmission using Block Cipher

A. Why does ECB reveal patterns?

ECB mode encrypts every plaintext block independently with the same secret key. Therefore, whenever two plaintext blocks are identical, their ciphertext blocks are also identical.

For financial applications, repeated values such as account numbers, transaction types, or customer records produce repeated ciphertext patterns. Although the attacker cannot immediately decrypt the information, these visible patterns reveal the structure of the original data and increase the possibility of cryptanalysis.

This makes ECB unsuitable for protecting confidential financial information.

B. Recommended mode

CBC (Cipher Block Chaining) is a more secure encryption mode.

Unlike ECB, CBC combines every plaintext block with the previous ciphertext block before encryption. The first block uses an Initialization Vector (IV), ensuring that encryption starts with a random value.

As a result:

- Repeated plaintext blocks generate completely different ciphertext.
- Patterns are hidden.
- Data confidentiality is significantly improved.

For financial systems where sensitive customer information is transmitted regularly, CBC provides stronger protection against pattern analysis attacks.

2. Choosing an Encryption Algorithm for Banking System

A. Suitable algorithm

The most appropriate algorithm is **Blowfish**.

B. Justification

Banking systems require encryption algorithms that provide both high security and efficient performance because they handle confidential customer information and process a large number of transactions every day.

DES is no longer recommended because its 56-bit key can be broken using brute-force attacks.

Although 3-DES offers improved security, it performs the encryption process three times, increasing computational overhead and response time.

Blowfish supports much larger key sizes and performs encryption quickly without sacrificing security. It provides a good balance between encryption strength and execution speed.

Therefore, Blowfish is well suited for banking applications where both security and system efficiency are equally important.

3. Key Exchange in an Unsecured Network

A. Explain the process

Diffie-Hellman enables two users to establish a common secret key over a public network without directly transmitting that secret key.

Initially, both users agree on public parameters. Each user generates a private key and calculates a corresponding public key. After exchanging the public keys, each user performs mathematical calculations using their own private key and the received public key.

Both users independently obtain the same shared secret key, which is later used for encrypting communication.

This method ensures that the secret key is never transmitted across the network.

B. Threat and mitigation

Despite its advantages, Diffie-Hellman is vulnerable to the **Man-in-the-Middle attack** because it does not authenticate the communicating users.

An attacker can intercept and replace public keys during transmission, establishing separate secure connections with each user.

To eliminate this vulnerability, authentication mechanisms such as Digital Certificates, Digital Signatures, and Public Key Infrastructure (PKI) should be implemented. These techniques verify the identity of communicating users and ensure the authenticity of exchanged public keys.

4. Public Key Cryptography for Email Security

A. RSA operation

RSA uses asymmetric cryptography based on public and private keys.

When an email is sent, the sender encrypts the message using the receiver's public key. Only the receiver's private key can decrypt the message, ensuring confidentiality.

Additionally, the sender may digitally sign the email using their private key. The receiver verifies the signature using the sender's public key, confirming message authenticity and integrity.

Thus, RSA provides secure communication by supporting confidentiality, authentication, integrity, and non-repudiation.

B. Importance of key management

The effectiveness of RSA depends largely on secure key management.

Organizations should distribute public keys through trusted Certificate Authorities while protecting private keys from unauthorized access. Regular key updates, secure storage, and certificate validation reduce the risk of compromised communication.

Effective key management forms the foundation of secure RSA-based email systems.

5. Cryptography for IoT Devices

A. Device constraints

IoT devices are designed to be compact, energy-efficient, and cost-effective. Consequently, they possess limited computational power, memory, storage capacity, and battery life.

Traditional encryption methods with large key sizes increase processor workload, memory usage, and energy consumption, reducing device performance and battery longevity.

B. Recommended approach

Elliptic Curve Cryptography (ECC) is the preferred cryptographic solution because it provides security equivalent to RSA while using significantly smaller keys.

Smaller keys reduce computational complexity, memory requirements, and power consumption. These advantages make ECC particularly suitable for smart home systems and other IoT environments where efficient resource utilization is essential.

Final Conclusion: ECC achieves an excellent balance between security and efficiency, making it the ideal cryptographic technique for modern low-power IoT devices.